

Full-Stack Coding Challenge: Word Cloud Generator

Goal: Create a full-stack application using **Node.js/TypeScript** with Express for the API, and **React** for data visualization.

Part 1: Backend API Server (Node.js/TypeScript & Express)

Task: Build a Node.js/TypeScript server using **Express** that processes the data and makes it available.

1. **Server Setup:** Create an **Express** server.
 2. **API Route:** Implement a single GET route that, when called, performs the following:
 - **API Fetch:** Call the URL `https://random-word-api.herokuapp.com/word?number=1` **6000 times**. (The query parameter `number=1` must remain unchanged for every request, ensuring each call retrieves a single word).
 - **Processing:** Calculate the **frequency** (count) of every unique word found.
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Part 2: Frontend Visualization (React)

Task: Build a React application to consume the API and visualize the data as an interactive Word Cloud.

1. **Data Retrieval:** The React application must **fetch** the processed word frequency data from the Express API.
2. **Visualization:** Display **every unique word** present in the data.
3. **Styling & Appearance:**
 - **Size Scaling:** The **font size** of each word must be **directly proportional** to its frequency. Use a **scaling function** to map the raw count to a practical font size range (e.g., 12px to 64px).
 - **Color:** Assign a **random, persistent color** to each unique word.
 - **Layout:** Arrange the words in a visually appealing "cloud" formation.

